

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 0.4548965096159157, median 0.3948269990307368, std: 0.2999333155695944
Gyroscope error (imu0): mean 1.0344073499184285e-09, median 1.1465681187882438e-10, std:
6.957416822054965e-09
Accelerometer error (imu0): mean 9.133131503704793e-09, median 7.49166462259838e-11, std:
1.8441548738930404e-07

Residuals

Reprojection error (cam0) [px]: mean 0.4548965096159157, median 0.3948269990307368, std: 0.2999333155695944
Gyroscope error (imu0) [rad/s]: mean 5.933800021320919e-13, median 6.577201842434782e-14, std:
3.991069871100959e-12
Accelerometer error (imu0) [m/s^2]: mean 6.470951484384938e-11, median 5.307949227540659e-13, std:
1.3066095362596242e-09

Transformation (cam0):

T_ci: (imu0 to cam0):

```
[[ 0.06164513 -0.99611432 0.06289779 0.00061617]
 [ 0.03958396 -0.06052811 -0.9973813 -0.00046105]
 [ 0.99731288 0.06397344 0.03569889 -0.00000021]
 [ 0.         0.         0.         1.         ]]
```

T_ic: (cam0 to imu0):

```
[[ 0.06164513 0.03958396 0.99731288 -0.00001953]
 [-0.99611432 -0.06052811 0.06397344 0.00058589]
 [ 0.06289779 -0.9973813 0.03569889 -0.00049859]
 [ 0.         0.         0.         1.         ]]
```

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
-0.16108667605408186

Gravity vector in target coords: [m/s^2]

[-4.24325632 0.87525364 -8.79756385]

Calibration configuration

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cam0

Camera model: pinhole

Focal length: [163.19620135256736, 155.3890609374247]

Principal point: [130.03377014752047, 159.9470617581387]

Distortion model: radtan

Distortion coefficients: [-0.03244175845474721, -0.011430444884891186, 0.010768262133809157, -0.024675241344505598]

Type: checkerboard

Rows

Count: 8

Distance: 0.02 [m]

Cols

Count: 11

Distance: 0.02 [m]

IMU configuration

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IMU0:

Model: calibrated

Update rate: 50.0

Accelerometer:

Noise density: 0.0010019901002151698

Noise density (discrete): 0.007085139945439349

Random walk: 1.9893785362909353e-05

Gyroscope:

Noise density: 8.112529814510266e-05

Noise density (discrete): 0.0005736424844418254

Random walk: 6.28721982800146e-07

T_ib (imu0 to imu0)

[[1. 0. 0. 0.]

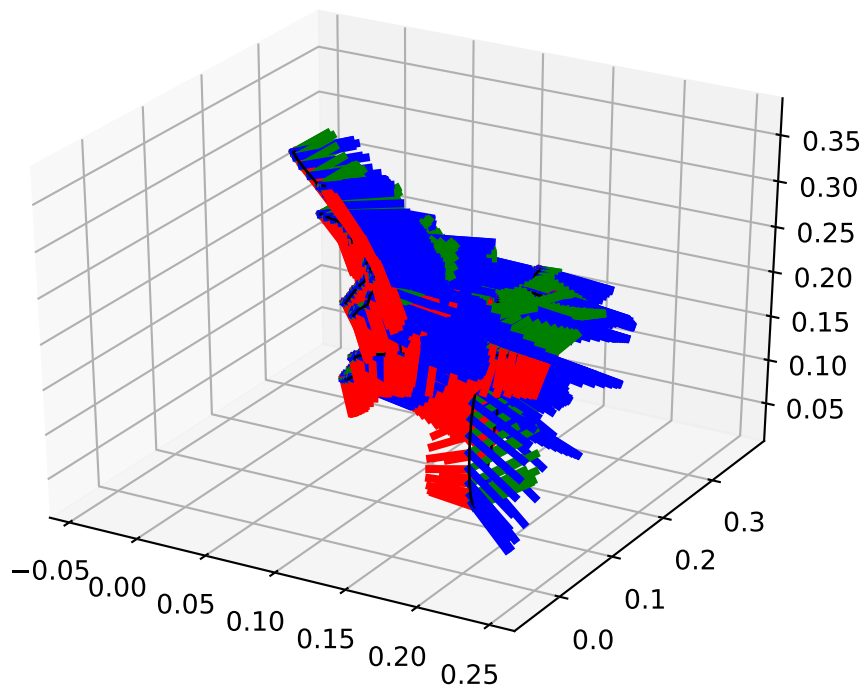
[0. 1. 0. 0.]

[0. 0. 1. 0.]

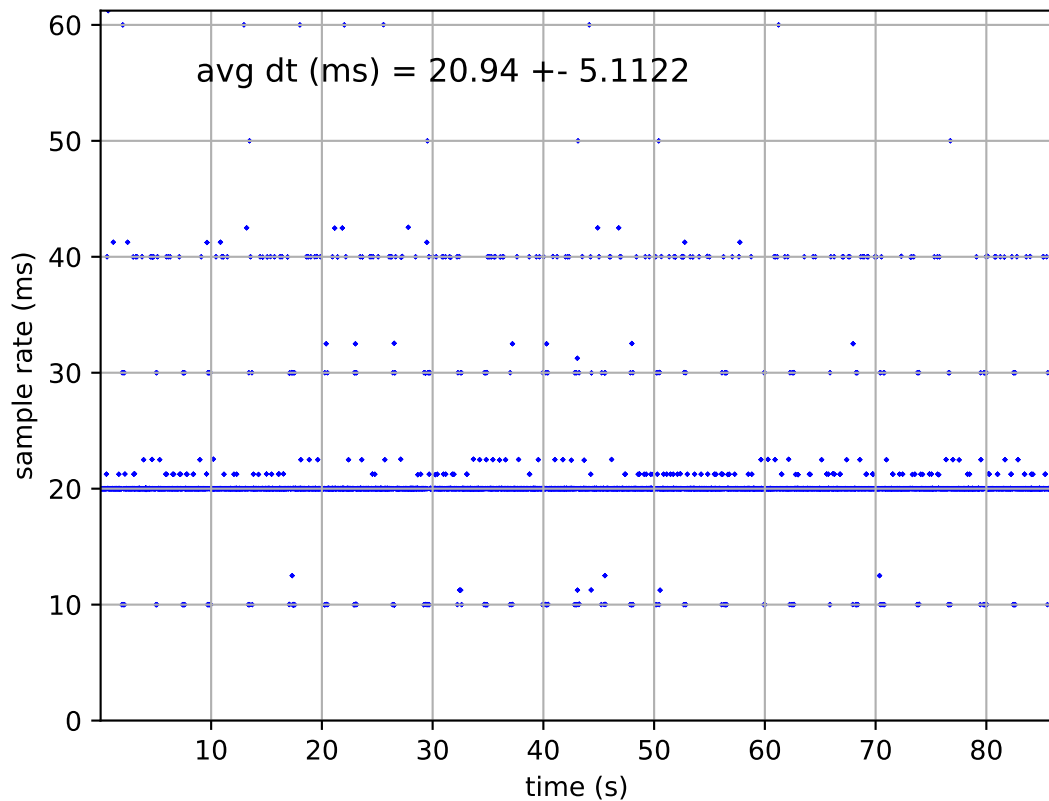
[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

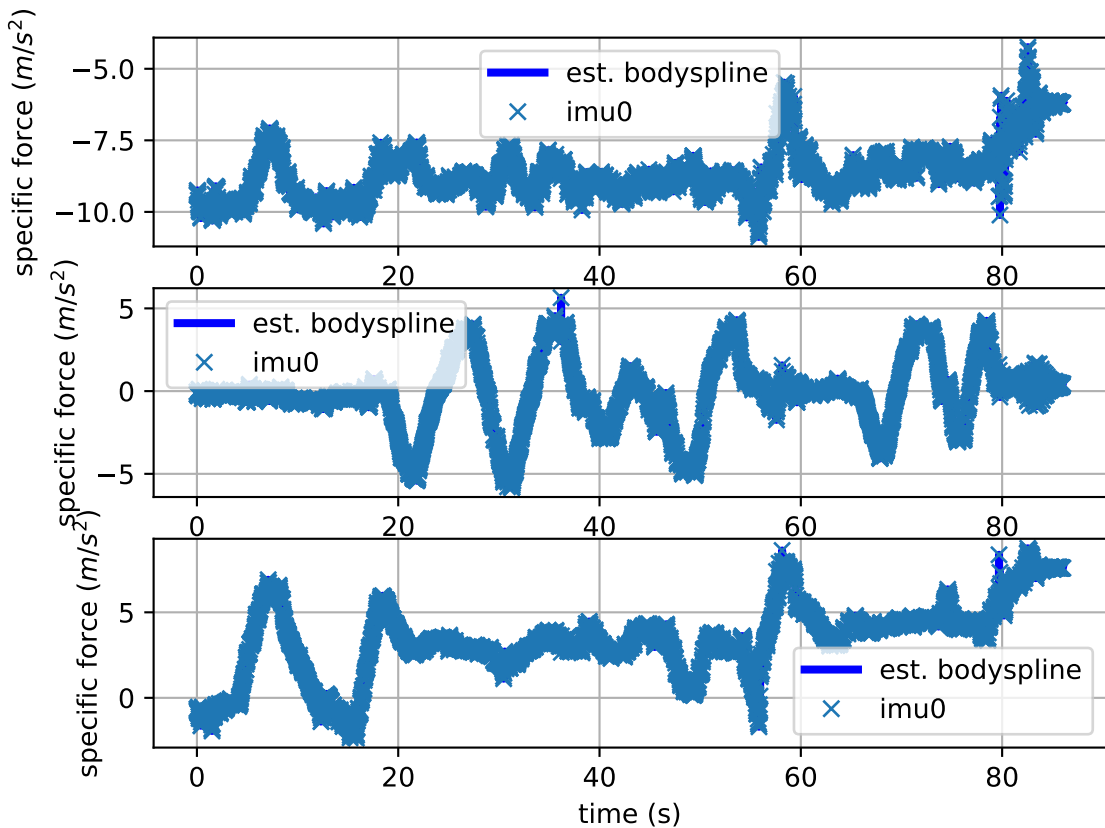
imu0: estimated poses



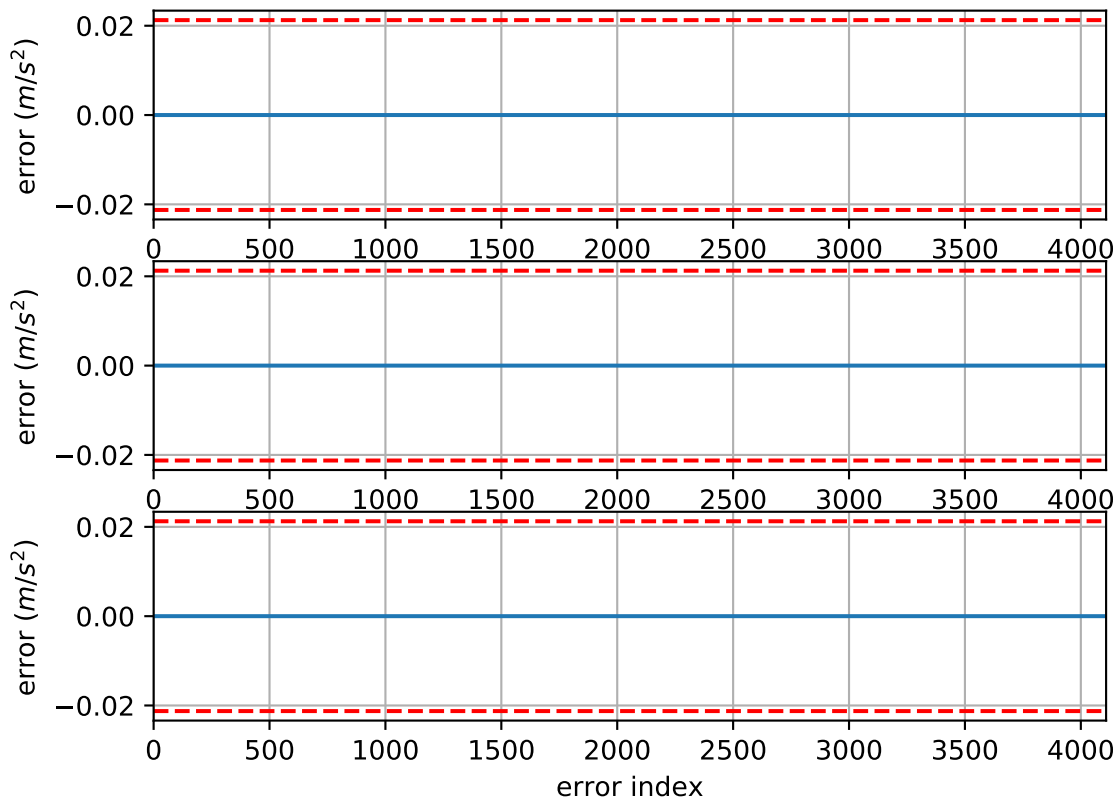
imu0: sample inertial rate



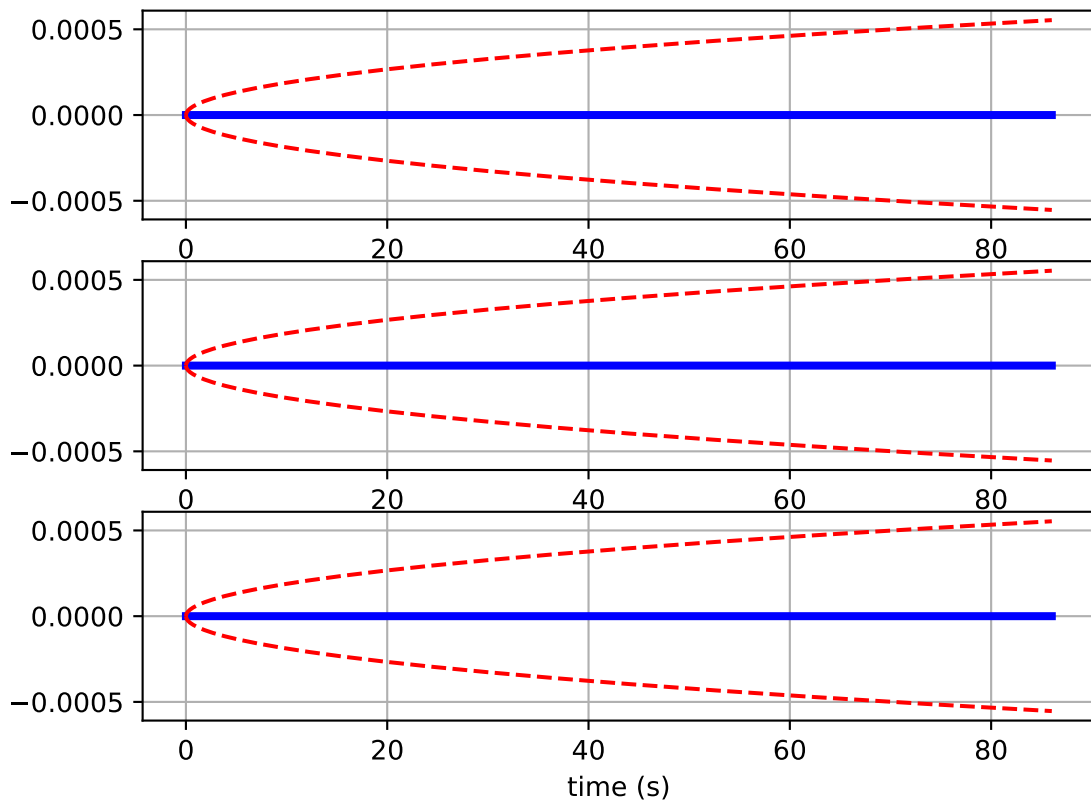
Comparison of predicted and measured specific force (imu0 frame)



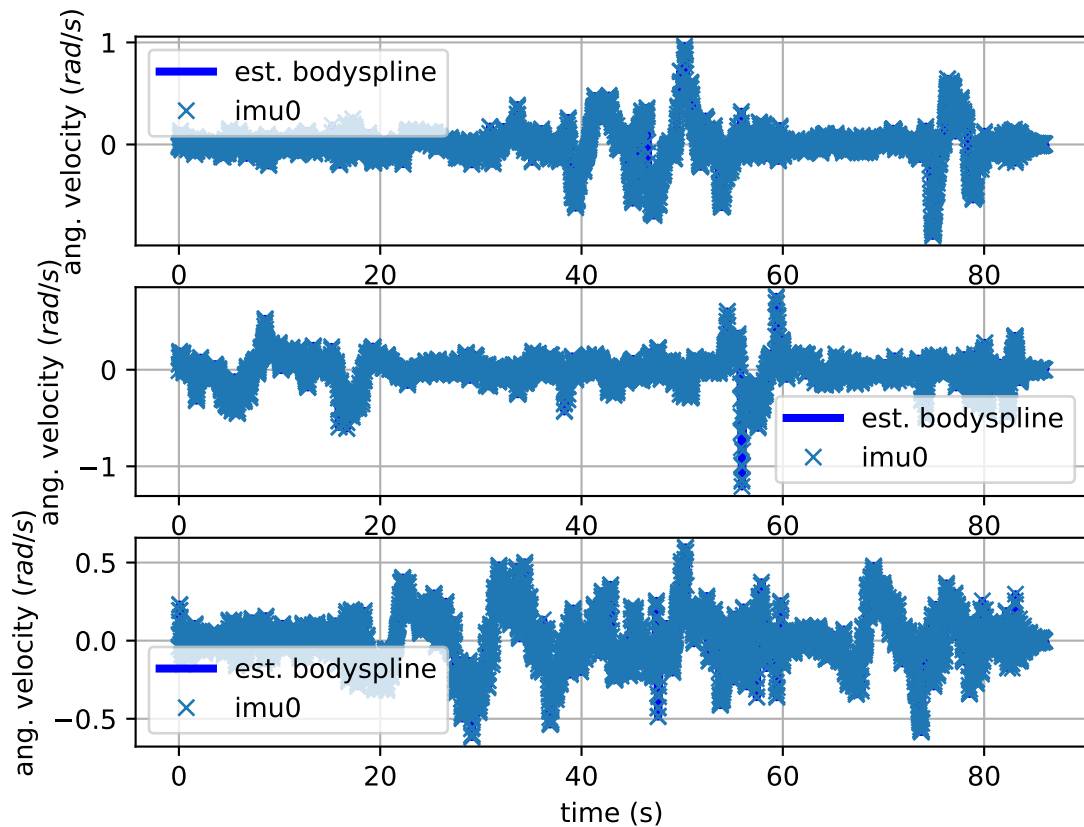
imu0: acceleration error



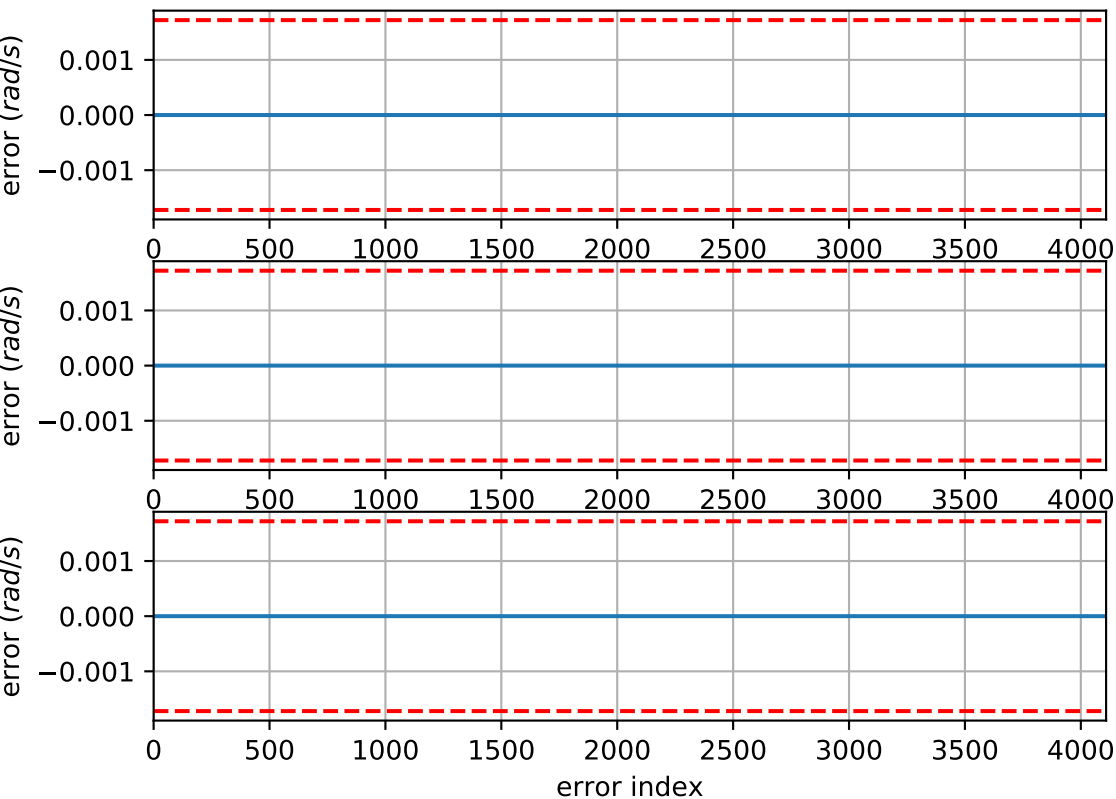
imu0: estimated accelerometer bias (imu frame)



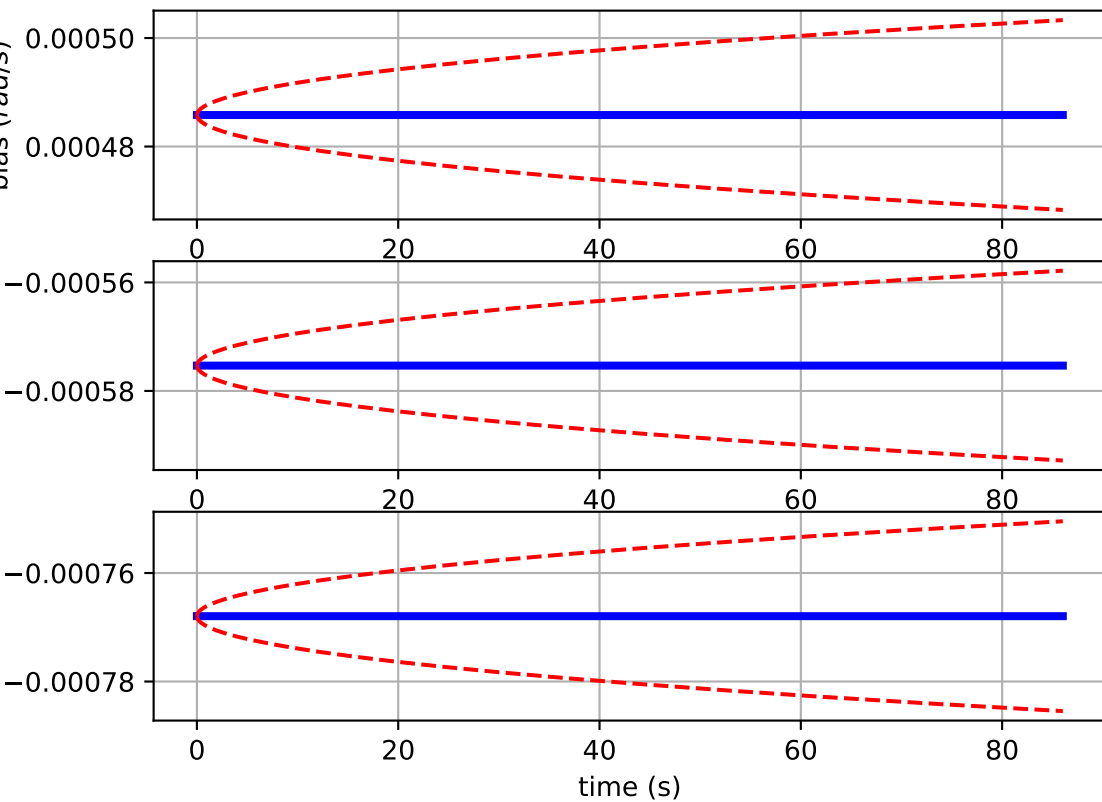
Comparison of predicted and measured angular velocities (body frame)



imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors

